
CS204: Algorithms & Data Structures Lab

Week # 10 (1 Questions, 103 Marks)

Lab session: AL1

Held on: 07-Oct-2024 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 4

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Question 1: (103 points)

Templates and Binary Search Tree.

Task 01 (5 marks) Create a class template `Node` with a `typename T` and with the following specifications

1. (5 marks) Public members
 - (a) (1 mark) A template to store an element within a node.
 - (b) (1 mark) A pointer to `Node` of type `T` representing left sub-tree
 - (c) (1 mark) A pointer to `Node` of type `T` representing right sub-tree
 - (d) (2 marks) A constructor with the following specification
 - i. Input argument 1 A specified `typename T`
 - ii. Functionality Initialize the data members appropriately.

Task 02 (56) Create a template class `BST` with a `typename T` and with the following specifications

1. Private data members and member functions
 - (a) (1 mark) Data member - A pointer to `Node` with template `typename T`
 - (b) Private Member functions -
 - i. (30 marks) `insert_node`
 - ii. (5 marks) `find_min`
 - iii. (5 marks) `find_max`
 - iv. (5 marks) `in_order_traversal`
 - v. (5 marks) `pre_order_traversal`
 - vi. (5 marks) `post_order_traversal`
 - (c) (30 marks) Specification of private member function `insert_node`
 - i. Input argument 1: A pointer to `Node` with template `typename T`
 - ii. Input argument 2: A data element of `typename T`
 - iii. Output A pointer to `Node` with `typename T`
 - iv. (25 marks) Functionality: This function inserts a new node with given value in the appropriate place of the tree.

- v. (5 marks) Throw exceptions of class object **Errors** in case of any errors with appropriate message.
- (d) Specification of private member function **find_min**
 - i. Input argument 1: A pointer to **Node** with template **typename T**
 - ii. Output A pointer to **Node** with **typename T**
 - iii. **Functionality:** This function find the minimum value in a given tree.
- (e) Specification of private member function **find_max**
 - i. Input argument 1: A pointer to **Node** with template **typename T**
 - ii. Output A pointer to **Node** with **typename T**
 - iii. **Functionality:** This function find the maximum value in a given tree.
- (f) Specification of private member function **in_order_traversal**
 - i. Input argument 1: A pointer to **Node** with template **typename T**
 - ii. Output **void**
 - iii. **Functionality:** Traverse the tree using **in order** traversal method and prints the elements of the tree.
- (g) Specification of private member function **pre_order_traversal**
 - i. Input argument 1: A pointer to **Node** with template **typename T**
 - ii. Output **void**
 - iii. **Functionality:** Traverse the tree using **pre order** traversal method and prints the elements of the tree.
- (h) Specification of private member function **post_order_traversal**
 - i. Input argument 1: A pointer to **Node** with template **typename T**
 - ii. Output **void**
 - iii. **Functionality:** Traverse the tree using **post order** traversal method and prints the elements of the tree.
- 2. The following public member functions
 - (a) Constructor
 - (b) **insert**
 - (c) **inorder**
 - (d) **preorder**
 - (e) **postorder**
 - (f) Public BST constructor with no arguments. The constructor should initialize the binary search tree

- (g) Specification of public member function **insert**
 - i. Input argument 1 A data element of **typename T**
 - ii. Output **void**
 - iii. **Functionality:** This function inserts a new node with given value in the appropriate place of the tree using the appropriate private member function defined above.
- (h) Specification of public member function **inorder**
 - i. Input argument 1 **void**
 - ii. Output **void**
 - iii. **Functionality:** This function traverses the tree using **in order** method using the appropriate private member function defined above.
- (i) Specification of public member function **preorder**
 - i. Input argument 1 **void**
 - ii. Output **void**
 - iii. **Functionality:** This function traverses the tree using **pre order** method using the appropriate private member function defined above.
- (j) Specification of public member function **postorder**
 - i. Input argument 1 **void**
 - ii. Output **void**
 - iii. **Functionality:** This function traverses the tree using **postin order** method using the appropriate private member function defined above.

Task 04 (2 marks) Declare **Errors** class with the following specifications

1. (1 mark) Private member - an in-built string object **errors**
2. (1 mark) Public member function - a constructor which initializes the data members of the class
3. You should throw errors of this class object using **try ... catch** exception handling. You should not throw library exceptions.

Task 05 (40 marks) Perform the following in the main function. You should ignore all **delete** statement lines in the input file.

1. (10 marks) Read the input file **int-bst.txt**
2. Perform the operations mentioned in the file

3. (10 marks) Read the input file `char-bst.txt`
4. Perform the operations mentioned in the file

5. (10 marks) Read the input file `float-bst.txt`
6. Perform the operations mentioned in the file

7. (10 marks) Read the input file `string-bst.txt`
8. Perform the operations mentioned in the file